Sina Shid-Moosavi

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Summary

Doctoral researcher with 5+ years of expertise in integrating machine learning and AI into complex engineering domains. Demonstrated proficiency in data analysis, interdisciplinary research, and translating theory into scalable engineering solutions.

Education

Northeastern University — Boston, MA
Ph.D. Candidate in Data and Systems — *2023 – Present*

University of Central Florida — Orlando, FL
M.Sc. in Smart Cities — *2021 – 2023*

Amirkabir University of Technology — Tehran, Iran
M.Sc. in Structural Engineering — *2014 – 2017*

Shiraz University — Shiraz, Iran
B.Sc. in Civil Engineering — *2010 – 2014*

Research Experience

Northeastern University — Boston, MA
Doctoral Researcher — *Sep. 2023 – Present*

- Addressing data scarcity using a Graph Neural Network that links turbines through spatial and wake-based edges.
- Built a Gaussian Process Regression model for wind turbine maintenance prediction using sensor data.
- Integrated physics into Gaussian Process Regression for wind, turbulence, and veer prediction with uncertainty.
- Applied statistical and machine learning methods to identify operational patterns and outliers in time-series data.
- Fused cepstral and image features for vibration-based damage detection, improving accuracy by 10%.
- Used Bayesian optimization to tune dynamic system parameters, cutting model runtime by 40%.
- Implemented Reinforcement Learning-based adaptive parameter tuning for real-time damage response.
- Performed sensitivity analysis to identify key drivers of wake-induced fatigue.
- Visualized large-scale data trends to support decision-making in model calibration.

University of Central Florida — Orlando, FL
Graduate Research Assistant — *Aug. 2021 – Aug. 2023*

- Implemented real-time passenger detection, achieving 98% accuracy on 4,300-image custom dataset.
- Used DeepSORT for multi-object tracking and OSNet for passenger re-identification with 92% Rank-1 accuracy.
- Deployed IoT-enabled Jetson TX2 edge-computing system for on-board real-time data processing and storage.
- Integrated location and video data to identify passenger origin-destination pairs for route-level analysis.
- Automated boarding and alighting detection across 12 bus stops with 82% accuracy for origin-destination analysis.
- Prepared a custom dataset with 3,060 images and 200 unique IDs for passenger re-identification.
- Fine-tuned models using a custom dataset to handle lighting and occlusion, improving accuracy by over 30%.
- Developed 3D deformation sensing via multi-camera photogrammetry for structural displacement quantification.
- Utilized UAV-based sensing to map surface water networks and estimate flooded areas in vegetated regions.

AmirKabir University of Technology — Tehran, Iran
Graduate Research Assistant — *Sep. 2014 – Feb. 2017*

- Investigated the seismic performance of integral and semi-integral pre-tensioned concrete bridges.
- Developed advanced Finite Element Analysis models to simulate bridge response under seismic loading conditions.
- Analyzed soil–structure interaction effects at bridge abutments and their influence on overall structural behavior.

Work Experience

Atmospheric G2 Manchester, NH
Data Science Intern *May. 2026 – Dec. 2026*

- Built neural network models to forecast regional wind power generation, achieving under 18% mean error.
- Fine-tuned models and improving forecast accuracy by 3–5% over the commercial forecasting products.
- Analyzed large-scale meteorological, generation, and consumption time-series datasets.
- Integrated weather signals into production forecasting pipelines with cross-functional teams.
- Evaluated model performance using diagnostics, error analysis, and bias assessment.
- Communicated insights through clear visualizations for technical and business stakeholders.

Miragar Tajheez Tehran, Iran
Structural Engineer *Sep. 2018 – Feb. 2021*

- Managed structural evaluation and rehabilitation projects to enhance seismic performance of existing buildings.
- Designed and detailed structural strengthening systems in compliance with national seismic and structural codes.
- Developed and implemented seismic control solutions to improve structural resilience under dynamic loading.
- Conducted destructive and nondestructive field testing to assess material properties and structural performance.
- Performed structural modeling and performance assessments to support retrofit and rehabilitation strategies.

Teaching Experience

Northeastern University **Boston, MA**

- Structural Analysis, Graduate Teaching Assistant *Fall 2025*
- Statics and Solid Mechanics, Graduate Teaching Assistant *Fall 2024*
- Structural Analysis, Graduate Teaching Assistant *Spring 2024*

University of Central Florida **Orlando, FA**

- Mechanics of Materials Laboratory, Graduate Teaching Assistant *Summer 2023*
- Mechanics of Materials Laboratory, Graduate Teaching Assistant *Spring 2023*
- Mechanics of Materials Laboratory, Graduate Teaching Assistant *Fall 2022*

Shiraz University **Shiraz, Iran**

- Statics, Undergraduate Teaching Assistant *Spring 2014*
- Strength of Materials I, Undergraduate Teaching Assistant *Spring 2014*
- Structural Analysis II, Undergraduate Teaching Assistant *Spring 2014*
- Structural Analysis I, Undergraduate Teaching Assistant *Fall 2013*
- Numerical Analysis, Undergraduate Teaching Assistant *Fall 2013*
- Structural Drawing, Undergraduate Teaching Assistant *Fall 2012*

Technical Skills

Programming Languages: Python, MATLAB
Python Libraries: TensorFlow, PyTorch, scikit-learn, NumPy, Pandas, GeoPandas, Xarray, Matplotlib, Seaborn
Machine Learning & Deep Learning: Decision Trees, Random Forest, SVM, KNN, ANN, CNNs, RNNs, LSTM, GNN, Transformers, R-CNN, FPN, U-Net, ResNet
Computer Vision: Image Classification, Object Detection, OpenCV, YOLO, OSNet, DeepSORT
Time-series & Sequential Modeling: ARIMA, LSTM, Forecasting, Anomaly Detection, Temporal Data Processing
Cloud and Computing: AWS, Linux (Ubuntu, Bash), Git
Engineering Software: OpenSees, ABAQUS, VecTor2, Floris, CSI Software (SAP2000, ETABS, SAFE, CSiBridge)
Geospatial Tools: ArcGIS, AutoCAD, DJI Terra, Agisoft Metashape, TrueView EVO/LP360
Scientific Writing: LaTeX (Overleaf)

Relevant Courses

Machine Learning and Pattern Recognition, Reinforcement Learning, Computer Vision, 3D Computer Vision, Image Processing, Time Series and Geospatial Data Sciences, and System Identification.

Publications

Journal Papers

1. Di Cioccio, F., **Shid-Moosavi, S.**, Haghi, R., Tronci, E., Moaveni, B., Liberatore, S., and Hines, E., (2026). "Calibration of Wake Models for the Block Island Offshore Wind Farm Using SCADA Measurements." *Journal of Fluids and Structures*, 104655.
2. **Shid-Moosavi, S.**, Di Cioccio, F., Haghi, R., Tronci, E., Moaveni, B., Liberatore, S., and Hines, E., (2025). "Modeling and Experimentally-Driven Sensitivity Analysis of Wake-Induced Power Loss in Offshore Wind Farms: Insights from Block Island Wind Farm." *Renewable Energy*, 122126.
3. **Shid-Moosavi, S.** and Rahai, A., (2018). "The Performance of Integral and Semi-integral Pre-tensioned Concrete Bridges under Seismic Loads in Comparison with Conventional Bridges." *Amirkabir Journal of Civil Engineering*, Volume 2, Issue 2, Page 219-226.

Conference Papers

1. **Shid-Moosavi, S.**, Tronci, E., and Caracoglia, L., (2027). "Quantifying the Impact of Physics Constraints on Uncertainty Propagation in Probabilistic Wake Modeling." *International Modal Analysis Conference (IMAC-XLV)*, Kissimmee, FL, USA. (*Upcoming: Jan 2027*)
2. Speciale, C., **Shid-Moosavi, S.**, Milana, S., and Tronci, E., (2026). "Dynamic Recognition of Damage States in Wind Turbines Using Cepstral Coefficients and ANOVA Analysis." *International Modal Analysis Conference (IMAC-XLIV)*, Palm Springs, CA, USA.
3. Tronci, E., **Shid-Moosavi, S.**, and Speciale, C., (2025). "Pattern Recognition and Damage Detection in Wind Turbine Monitoring Using Cepstral Coefficients." *11th International Conference on Experimental Vibration Analysis of Civil Engineering Structures (EVACES 2025)*, Porto, Portugal.
4. **Shid-Moosavi, S.**, Speciale, C., and Tronci, E., (2025). "Damage Identification Strategy in Time-Varying Dynamic Systems Combining Cepstral and Image-Based Features." *International Modal Analysis Conference (IMAC-XLIII)*, Orlando, FL, USA.
5. **Shid-Moosavi, S.**, Hassan, Z., and Sun, P., (2022). "Towards Full-field Sensing of 3D Deformation in Structural Components using Multi-camera Photogrammetry." *8th World Conference on Structural Control and Monitoring (8WCSCM)*, Orlando, FL, USA.
6. Vasef, M., Marefat, M., **Shid-Moosavi, S.**, and Sun, P., (2022). "Monitoring the Seismic Behavior of a Scaled RC Frame of Intermediate Ductility in a Shaking Table Test." *8th World Conference on Structural Control and Monitoring (8WCSCM)*, Orlando, FL, USA.
7. Abasi, A., **Shid-Moosavi, S.**, and Rahai, A., (2019). "Seismic performance of integral, semi-integral, and conventional bridges." *5th International Conference on Bridge (5IBC2019)*, Tehran, Iran.
8. **Shid-Moosavi, S.** and Rahai, A., (2019). "The performance of integral and semi-integral pre-tensioned concrete bridges under thermal loads depending on various deck lengths and joint conditions." *11th National Congress on Civil Engineering (11-NCCE)*, Shiraz, Iran.

Presentations

Oral Presentations

1. **Shid-Moosavi, S.**, Tronci, E., Moaveni, B., and Hines, E., (2027). "Comparative Assessment of Fore-Aft and Side-to-Side Fatigue Damage in a 6 MW Offshore Wind Turbine Tower." *International Modal Analysis Conference (IMAC-XLV)*, Kissimmee, FL, USA. (*Upcoming: Jan 2027*)
2. **Shid-Moosavi, S.**, Tronci, E., Hines, E.,, and Moaveni, B., (2026). "A Two-Step Gaussian Process Framework for Turbulence Intensity Reconstruction and Fatigue Estimation." *North American Wind Energy Academy Conference (NAWEA/WindTech 2026)*, Portland, OR, USA. (*Upcoming: Sep 2026*)
3. **Shid-Moosavi, S.**, Speciale, C., Tronci, and Milana, S., (2026). "Reinforcement Learning-Supervised Framework for Robust Damage Detection in Wind Turbines." *North American Wind Energy Academy Conference (NAWEA/WindTech 2026)*, Portland, OR, USA. (*Upcoming: Sep 2026*)
4. **Shid-Moosavi, S.**, Tronci, E., Moaveni, B., and Hines, E., (2026). "Graph Neural Network Modeling of Turbulence-Induced Fatigue in Offshore Wind Turbine Towers." *International Modal Analysis Conference (IMAC-XLIV)*, Palm Springs, CA, USA.
5. **Shid-Moosavi, S.**, Speciale, C., Milana, S., and Tronci, E., (2026). "Reinforcement Learning-Based Tuning of Damage Detection Parameters for Wind Turbine Diagnostics Using Cepstral Coefficients." *International Modal Analysis Conference (IMAC-XLIV)*, Palm Springs, CA, USA.

6. **Shid-Moosavi, S.**, Tronci, E., Hines, E., Kirincich, A., and Moaveni, B., (2025). "Data Driven Mapping of Turbulence Intensity for Wake Modeling of Offshore Wind Systems." *North American Wind Energy Academy Conference (NAWEA/WindTech 2025)*, Dallas, TX, USA.
7. Di Cioccio, F., **Shid-Moosavi, S.**, Haghi, R., Tronci, E., Moaveni, B., Liberatore, S., and Hines, E., (2025). "Calibration of Wake Model in a Wind Farm using Measured Data: A Case Study of Block Island Wind Farm." *Wind Energy Science Conference 2025 (WESC 2025)*, Nantes, France.
8. **Shid-Moosavi, S.**, Di Cioccio, F., Haghi, R., Tronci, E., Moaveni, B., Liberatore, S., and Hines, E., (2025). "Investigating the Impact of Wake-Induced Turbulence on Tower Fatigue Accumulation in the Block Island Wind Farm." *The First New England Fluid-Structure Interactions Symposium (NEFSI 25)*, Amherst, MA, USA.
9. Di Cioccio, F., **Shid-Moosavi, S.**, Haghi, R., Tronci, E., Moaveni, B., Liberatore, S., and Hines, E., (2025). "Model Updating for Improved Wake Prediction: Balancing Fidelity and Computational Cost in Offshore Wind Farms." *The First New England Fluid-Structure Interactions Symposium (NEFSI 25)*, Amherst, MA, USA.
10. **Shid-Moosavi, S.**, Di Cioccio, F., Haghi, R., Tronci, E., Moaveni, B., Liberatore, S., and Hines, E., (2025). "Assessing Wake-Induced Fatigue in Offshore Wind Turbines Using Dynamic and SCADA Data." *International Modal Analysis Conference (IMAC-XLIII)*, Orlando, FL, USA.
11. **Shid-Moosavi, S.**, Partovi Mehr, N., Tronci, E., Moaveni, B., and Hines, E., (2024). "Pattern Recognition in Offshore Wind Turbine Dynamics: Unveiling Fatigue and Damage Signatures." *Engineering Mechanics Institute Conference and Probabilistic Mechanics & Reliability (EMI/PMC 2024)*, Chicago, IL, USA.
12. Iraniparast, M., **Shid-Moosavi, S.**, Sun, P., Wang, T., Apostolakis, G., and Mackie, K., (2023). "Multi-vision System for Full-field Strain Measurement and Crack Tracking on UHPC Beams." *Engineering Mechanics Institute Conference (EMI 2023)*, Atlanta, GA, USA.
13. **Shid-Moosavi, S.**, Sun, P., Jariz, C., and Wang, D., (2022). "UAV-based Remote Sensing to Measure Water Surface Areas in Florida." *AGU 2022 Conference*, Chicago, IL, USA.
14. Sun, P., **Shid-Moosavi, S.**, Jariz, C., Kanya, A., and Wang, D., (2022). "UAV-based watershed surveying over Florida forests." *FL-ASPRS/UF Spring 2022 LiDAR Workshop*, Orlando, FL, USA.

Poster Presentations

1. **Shid-Moosavi, S.**, Tronci, E., Moaveni, B., and Hines, E., (2026). "Linking Turbulence to Structural Fatigue in Offshore Wind Turbines via Data-Driven Modeling." *The Second New England Fluid-Structure Interactions Symposium (NEFSI 26)*, Boston, MA, USA.
2. **Shid-Moosavi, S.**, Tronci, E., and Caracoglia, L., (2025). "Physics-Informed Gaussian Process Framework for Offshore Wind Profile Reconstruction and Wake Prediction." *North American Wind Energy Academy Conference (NAWEA/WindTech 2025)*, Dallas, TX, USA.
3. **Shid-Moosavi, S.**, Di Cioccio, F., Haghi, R., Tronci, E., Moaveni, B., Liberatore, S., and Hines, E., (2024). "Experimentally Driven Sensitivity Analysis of Operational Parameters for Wake-induced Power Loss in the Block Island Offshore Wind Farm." *North American Wind Energy Academy Conference (NAWEA/WindTech 2024)*, New Brunswick, NJ, USA.

Honors and Academic Achievements

Department of Civil and Environmental Engineering PhD Teaching Award, Northeastern University	*2026*
Best Poster Award, CEE Graduate Research Exposition, Northeastern University	*2024*
Northeastern University CEE PhD Fellowship Recipient	*2023*
University of Central Florida ORCGS Fellowship Recipient	*2021*
Ranked 7th among approximately 3000 participants in the nationwide university entrance exam in Structural Engineering for the Ph.D. Degree	*2018*
Ranked 6th among 30 students in master's program at AmirKabir University of Technology	*2017*
Merit-based admission to M.Sc. Structural Engineering program at AmirKabir University of Technology without the need to take the university entrance exam due to the high GPA as a talented student	*2014*
Ranked 3rd among 50 students in undergraduate studies at Shiraz University	*2014*
2nd-ranked student award among the graduated students of Civil and Environmental Engineering Department of Shiraz University based on GPA and academic activities	*2014*
Ranked among the top 1% of students out of 500,000 participants in the National University Entrance Exam	*2010*
Awarded to study in National Organization for Development of Exceptional Talents (NODET) for High School and Middle School	*2006* *2003*

Professional Leadership

Northeastern University Boston, MA

Graduate Student Leadership Roles *Sep. 2024 – Present*

- Department of CEE Representative, COE PhD Council
- Vice President, CEE Graduate Student Council (GSC)
- President, Northeastern Graduate Structural Engineering Association (NGSEA)
- Vice President, Northeastern Graduate Structural Engineering Association (NGSEA)

References

References available upon request.